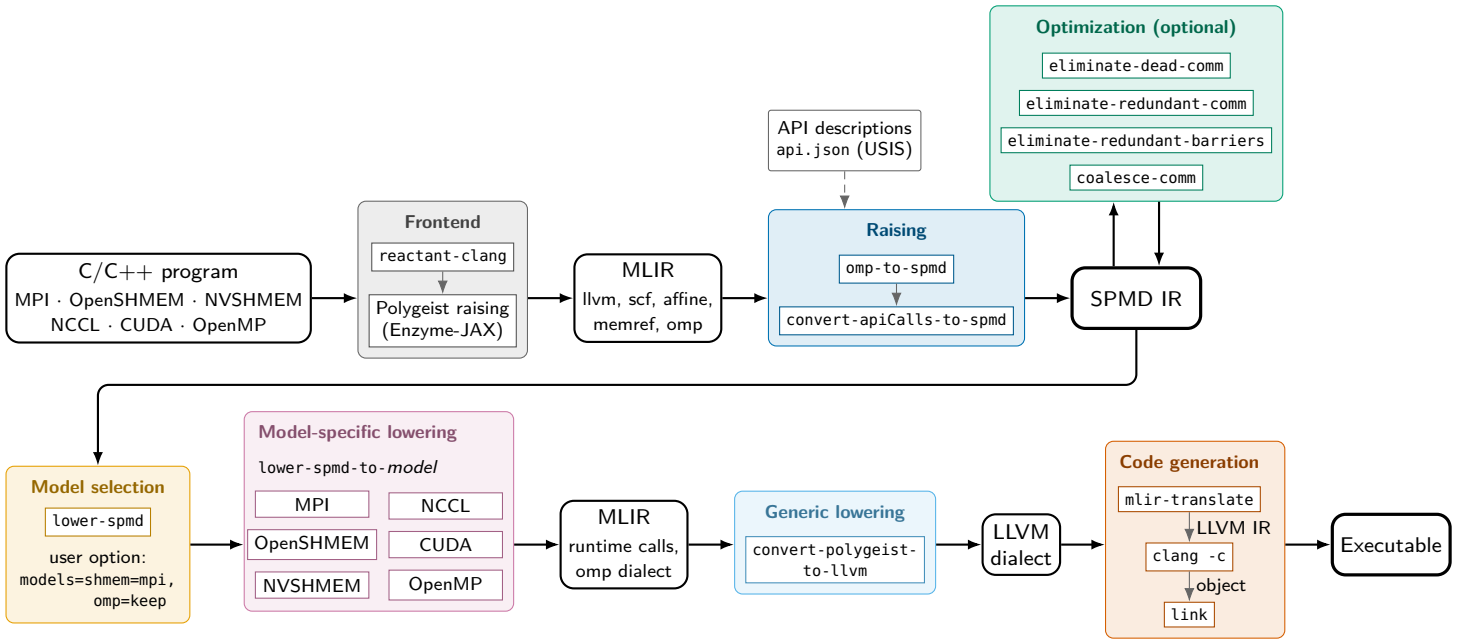

C/C++ program
MPI · OpenSHMEM · NVSHMEM
NCCL · CUDA · OpenMP
Frontend
reactant-clang
Polygeist raising
(Enzyme-JAX)
MLIR
llvm, scf, affine,
memref, omp
API descriptions
api.json (USIS)
Raising
omp-to-spmd
convert-apiCalls-to-spmd
Optimization (optional)
eliminate-dead-comm
eliminate-redundant-comm
eliminate-redundant-barriers
coalesce-comm
SPMD IR
Model selection
lower-spmd
user option:
models=shmem=mpi,
omp=keep
Model-specific lowering
lower-spmd-to-model
MPI
NCCL
OpenSHMEM
CUDA
NVSHMEM
OpenMP
MLIR
runtime calls,
omp dialect
Generic lowering
convert-polygeist-
to-llvm
LLVM
dialect
Code generation
mlir-translate
LLVM IR
clang -c
object
link
Executable